STAT 361 Assignment 1 Question 3

Edward Tanurkov

2026-09-24

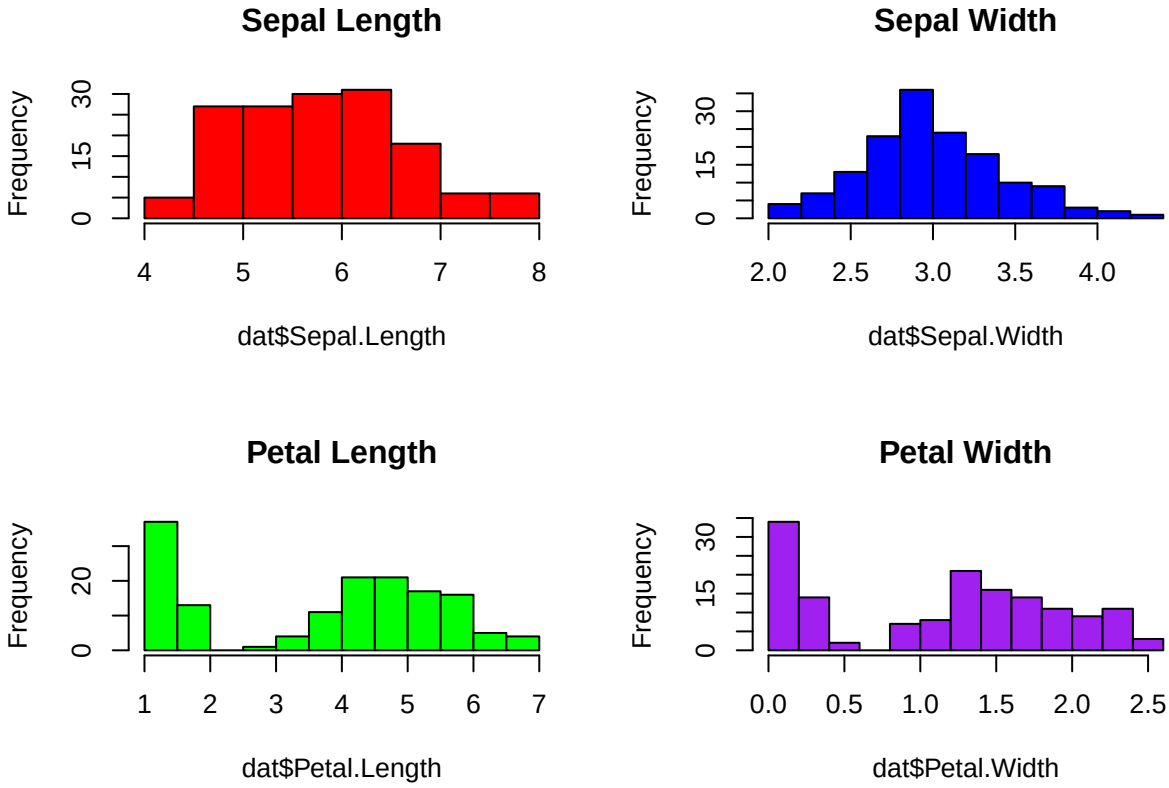
Load data

```
dat = iris
head(dat)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1          5.1         3.5          1.4         0.2  setosa
## 2          4.9         3.0          1.4         0.2  setosa
## 3          4.7         3.2          1.3         0.2  setosa
## 4          4.6         3.1          1.5         0.2  setosa
## 5          5.0         3.6          1.4         0.2  setosa
## 6          5.4         3.9          1.7         0.4  setosa
```

Plot data

```
par(mfrow = c(2, 2)) # Set 2x2 grid
hist(dat$Sepal.Length, col = "red", main = "Sepal Length")
hist(dat$Sepal.Width, col = "blue", main = "Sepal Width")
hist(dat$Petal.Length, col = "green", main = "Petal Length")
hist(dat$Petal.Width, col = "purple", main = "Petal Width")
```



Sepal Length plot: Seems like a slightly skewed Gaussian (normal) distribution with a mean around 5.5, but overall a pretty even distribution among the variables. Seems very feasible for linear regression.

Sepal Width plot: Much more prominent Gaussian, mean around or just below 3.0. Seems very feasible for linear regression.

Petal Length and Width plots: Bimodal distribution present in each, so predicting anything via a mean would be quite difficult given it will be skewed dramatically. Perhaps using a median-based residual function would be more appropriate, or using a different method to linear regression entirely. This bimodality makes it suitable for a classification problem as well.